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How is Labor Demand Changing as AI Improves? New Global Evidence

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ABSTRACT: The rapid advance of generative artificial intelligence (AI) has sparked intense debate about its effects on labor markets. We draw on over 700 million online job postings to present new evidence covering 110 economies. Demand has declined by nearly 10 percent in more AI-exposed occupations but, surprisingly, this impact has stabilized even as AI has become more capable. To understand this pattern, we develop a novel measure of AI adoption based on references to emerging AI skills in job postings. Demand for these skills has grown rapidly, particularly in more exposed occupations. Indeed, firms with stronger demand for AI skills exhibit faster growth in job postings, even in more exposed occupations. AI appears to be reshaping labor demand by both compressing hiring in more exposed occupations and generating new opportunities for workers with AI skills.

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1 Introduction

The rapid advance of generative artificial intelligence (AI) has generated widespread interest among economists and policymakers in its potential to replace workers and cut jobs. Both the media and AI company executives have predicted a bleak outlook for employment.[1] A growing body of work examines the impact of AI on labor demand (Brynjolfsson et al., 2026; Jaumotte et al., 2026; Liu et al., 2025; Hampole et al., 2025; Liu and Webber, 2026; Huang et al., 2026; Lambert and Schindler, 2026; Gimbel et al., 2026). This work generally relies on data for the United States and Europe, has yet to reach consensus, and importantly, has not focused on improvements in AI capabilities over time.

We use more than 700 million online job postings across 110 economies to provide new global evidence on how AI is reshaping labor demand. Using a standard difference-in-differences design, we find that postings have declined by nearly 10 percent in more AI-exposed occupations relative to less exposed occupations. Strikingly, the effect has not worsened even as AI has become increasingly capable at handling complex tasks. To our knowledge, we are the first to document this pattern. To understand why, we develop a new measure of AI adoption based on references to emerging AI skills in job postings. The measure allows us to track AI adoption across economies, industries, occupations, firms, and time.[2] We show that demand for emerging AI skills is rising and is concentrated in occupations with higher AI exposure. Moreover, job postings are growing faster at firms with stronger demand for AI skills, even in more exposed occupations.

We draw on a global dataset of 700 million online job postings across a broad range of economies, industries, and occupations from November 2021 through April 2026. The data is from Lightcast, a provider that compiles vacancies scraped from major recruitment

[1] For example, Forbes in June 2022 published an article titled "As AI Advances, Will Human Workers Disappear?" Bloomberg in August 2024 wrote about the impact of AI on call centers in the Philippines, and Axios in May 2025 and CNBC in October 2025 discussed the potential devastation for white-collar jobs.

[2] We are releasing economy-month and occupation-level AI adoption indices as supplementary materials accompanying this paper.

platforms.[3] The data contain detailed information on employers, occupations, industries, required skills, and in some cases, minimum experience requirements and salary. We complement these data with a measure of AI exposure from Pizzinelli et al. (2023), as well as frontier AI capabilities from METR, a nonprofit research organization that benchmarks progress in frontier AI models (METR, 2026).

We adopt a standard difference-in-differences specification to investigate how AI has affected labor demand for more exposed occupations relative to less exposed occupations, before and after the launch of ChatGPT. In our baseline results, we identify occupations that are more exposed to AI as in Pizzinelli et al. (2023). Importantly, we use granular fixed effects to restrict attention to comparisons across occupations within the same economy, industry, and month. This helps us address concerns that any observed decline in labor demand is driven by general sectoral trends, not the rise of AI (Gimbel et al., 2026). We also explicitly allow for differential trends in occupations more amenable to remote work, as previous papers argue that much of the observed negative impact of AI on more-exposed occupation vanishes when remote-workability is taken into account (Lambert and Schindler, 2026). The key identifying assumption is that absent the rise of generative AI beginning with ChatGPT's release in November 2022 trends in labor demand would have evolved similarly for more- and less-exposed occupations.

Our first finding is that after ChatGPT was released in November 2022, highly exposed occupations experienced a nearly 10 percent decline in job postings relative to less exposed occupations.[4] After our granular fixed effects and controls, we do not find any evidence of differential trends prior to the launch of AI. The impact of AI on labor demand was initially small, but emerged starting in late 2023. Our results are robust to a range of alternative specifications, including different measures of labor demand, different definitions

[3]Due to the nature of the data, our findings primarily reflect adjustments in formal-sector labor demand and, particularly in emerging and developing economies, may understate responses occurring through informal employment or channels that do not rely on online job postings.

[4]Note that postings capture demand for new hiring, not the existing stock of jobs. The full impact on the workforce will accumulate over time.

of AI exposure, and restricting the sample of economies to those with stronger Lightcast data coverage.[5] Across these specifications, the qualitative patterns remain consistent with the baseline results.

Indeed, since November 2022, dozens more AI models have been released, each one increasingly capable at handling complex tasks and processes. We use the high frequency nature of our job postings data to tie shifts in labor demand at each point in time to AI capabilities. Specifically, we link the job postings data to milestones in AI capabilities compiled by METR (2026). For this analysis, we group months according to the AI capability milestone prevailing at that point in time, rather than indexing event time by calendar months. Strikingly, the evidence to date shows that the impact of AI on labor demand has not worsened as AI capabilities improved dramatically.

To understand why the employment effects of AI have stabilized, we develop a novel measure of AI adoption based on references to emerging AI skills in job postings. Previous work emphasizes that technological change gives rise to new forms of work (Autor et al., 2026; Acemoglu et al., 2022; Atalay et al., 2020; Jaumotte et al., 2026). Motivated by this evidence, we identify AI-related skills that were rare in the early 2010s but experienced substantial growth after the arrival of generative AI, following Atalay et al. (2020) and Jaumotte et al. (2026). We combine the set of skills we identify with skills that have similar temporal patterns in existing taxonomies from Jaumotte et al. (2026) and Lightcast to construct a final set of 51 emerging AI skills.[6] These skills focus on general familiarity with AI (e.g., knowledge of ChatGPT or Copilot), expertise with AI (e.g., prompt engineering or AI agents), and the ability to connect AI with proprietary internal data and knowledge (e.g., vector databases and retrieval augmented generation). We use postings that ask for these skills to construct a new measure of AI adoption across economies, industries, occupations, and firms.

We find that demand for these skills has increased rapidly since 2023 and is concentrated

[5]The quality of coverage varies significantly across economies. We err on including a broad set of economies in the interest of presenting global evidence, but we find it reassuring that the results remain similar when focusing on economies with the best coverage.

[6]These existing taxonomies capture both generative AI and previous waves of AI.

in occupations with higher AI exposure. This validates the measure of exposure constructed by Pizzinelli et al. (2023), which we use in our difference-in-differences specification: the occupations hypothesized to be more exposed to AI are indeed the ones where AI skills are more likely to be in demand. Furthermore, these patterns suggest that firms are increasingly adapting production processes and job tasks to incorporate AI, resulting in a reallocation of labor demand rather than continued reductions in hiring. And because these skill shifts are concentrated in highly exposed occupations, they strengthen the interpretation that the effects identified in our difference-in-differences framework are driven by AI rather than coincident labor market trends.

Evidence at the firm level helps sharpen this picture of growing demand for emerging AI skills. For this analysis, we use our novel measure of AI adoption to identify firms that are AI adopters and compare their posting trends to non-AI adopting firms. We find that firms with stronger demand for AI skills exhibit faster growth in job postings, and this holds true even for occupations that are more exposed to AI.[7] This firm-level evidence is consistent with Aghion et al. (2025), who report similar patterns for French firms for adoption of AI prior to the rise of generative AI. Our analysis shows that these patterns generalize across a much broader set of economies and firms in the era of generative AI.

Taken together, our findings suggest that firms are increasingly adapting production processes and job tasks to incorporate AI, resulting in a reallocation of labor demand rather than deeper reductions in hiring. Generative AI is reshaping labor demand by both compressing hiring in more exposed occupations and generating new opportunities for workers with AI skills.

This paper contributes to three strands of literature. First, we add to the rapidly growing body of work studying the impact of AI on labor demand by presenting the first evidence linking shifts in demand with improvements in AI capabilities over time. Early studies in

[7]Importantly, hiring tends to be more concentrated in high-exposure occupations for firms that adopt AI per our measure. As with our baseline difference-in-differences evidence, our findings at the firm level are robust to alternative samples and measures of AI exposure.

specific settings such as online freelance platforms found significant negative effects on the number of jobs for more exposed occupations relative to less exposed occupations (Hui et al., 2024; Demirci et al., 2025).[8] More recent studies that focus on broader settings via payroll and job postings data have focused primarily on the United States and Europe and have not yet reached consensus on the sign or magnitude of the effect (Brynjolfsson et al., 2026; Liu et al., 2025; Klein Teeselink, 2025; Hampole et al., 2025; Hosseini Maasoum and Lichtinger, 2025; Johnston and Makridis, 2025; Humlum and Vestergaard, 2025; Liu and Webber, 2026; Ahn and Carollo, 2026). Jaumotte et al. (2026) present evidence at the commuting zone level for the United States using Lightcast to date when AI skills "enter" each market and labor-market data from American Community Survey to capture zone-wide outcomes, finding negative employment effects following the entry of AI skills.[9] Our work presents some of the first global evidence for the impact of AI on employment, showing that AI reduced hiring in more-exposed occupations.[10] To our knowledge, prior work has not linked changes in labor demand to advancements in AI capabilities over time.

Second, we contribute to a burgeoning literature tracking adoption of generative AI. Existing measures rely primarily on surveys, platform-specific usage, or worker resume data. Bick et al. (2026a) use worker and firm surveys for the US and six European economies and document a large gap in AI adoption between US and Europe. Similarly, Yotzov et al. (2026) field surveys of business executives to measure firm-level AI adoption in the US, United Kingdom, Germany, and Australia. Using real-time survey data from the U.S. Business Trends and Outlook Survey to develop a measure of firm AI adoption, Bonney et al. (2024) document a rapid increase in AI use across firms between 2023 and 2024. Dias da Silva et al. (2026) draw on the ECB's Consumer Expectations Survey panel to document a significant

[8]Earlier work used measures of exposure and preparedness to adjust to AI to quantify the potential impact of AI on labor markets (Cazzaniga et al., 2024) and more broadly (Cerutti et al., 2025).

[9]Jaumotte et al. (2026) also present evidence for Germany drawing on data from Integrated Employment Biographies.

[10]In contemporaneous work, Huang et al. (2026) examine cross-country evidence and also find negative impact, primarily in high-income economies, and Chandar and Klein Teeselink (2026) consider a sample of 41 countries and find that AI-adopting firms reduce senior employment.

increase in AI use in Europe.[11] Researchers at Anthropic, Google, OpenAI, and Microsoft use their platform usage data to track AI adoption (Appel et al., 2026; Iscenko et al., 2026; Chatterji et al., 2026; Misra et al., 2025), while researchers at Ramp link firms' spending on AI to Revelio Labs workforce records for a set of firms in the United States (Kharazian et al., 2026).[12] Hampole et al. (2025) measure firm-level AI adoption by using Revelio Labs resume data to identify employees who develop AI applications and identify task level exposure to AI applications. Bick et al. (2026b) use the Real-Time Population Survey to measure AI adoption at the occupation and task level for the U.S. By identifying emerging AI skills directly from employers' hiring requirements, we construct a complementary measure of adoption that can be tracked consistently across a broad set of economies, industries, occupations, and firms. This also allows us to present evidence on how labor demand is shifting differently at firms that adopt generative AI.

Finally, we connect to the literature studying how technological change gives rise to new kinds of work. Autor et al. (2026) show that technological change creates "new work" requiring novel expertise, and that workers who perform these new tasks earn wage premiums that reflect scarce skills, helping offset automation-driven labor displacement and sustain labor demand over time. Acemoglu et al. (2022) show that establishments exposed to AI grow AI hiring, reduce non-AI hiring, and change skill requirements in their postings. Aghion et al. (2025) show that employment and sales rise for French firms after AI adoption. Atalay et al. (2020) document that technological change reshapes work not only through shifts across occupations but also through substantial changes in the tasks performed within existing occupations. Jaumotte et al. (2026) document the rapid growth of new skills, especially in AI and IT, and show that their diffusion creates new labor market opportunities while reshaping the demand for workers across occupations. Building on this literature, we identify AI-

[11] Abel et al. (2026) use business surveys and find increasing AI adoption in New York and surrounding states. Dodini and Smith (2026) present evidence on job postings in Texas. Bonin et al. (2026) use Danish firm-level data to study AI adoption and firm size.

[12] Earlier work by Acemoglu et al. (2022) uses job postings data from Lightcast for the United States to study the rise of AI skills prior to the rise of generative AI from 2010 to 2018.

related skills that emerged following the advent of generative AI, document their increasing prevalence in more AI-exposed occupations, and show that hiring is growing faster at firms that have adopted generative AI.

The rest of the paper is organized as follows. Section 2 describes our data sources and provides descriptive statistics about our sample. Section 3 analyzes the differential impact of AI on labor demand for more exposed occupations relative to less exposed occupations. Section 4 proposes a new measure of AI adoption and studies trends in emerging AI skills for occupations and firms. Section 5 concludes.

2 Data

This section describes the data sources and key measures we use in our analysis. We construct a global panel of approximately 700 million job postings from Lightcast covering 110 economies between November 2021 and April 2026, and use the rich information contained in these postings to measure labor demand, skill requirements, and AI adoption. We combine these data with occupation-level measures of AI exposure based on the complementarity-adjusted AI Occupational Exposure (C-AIOE) measure constructed by Pizzinelli et al. (2023) and with external data on frontier AI capability milestones (METR, 2026). Together, these datasets allow us to examine how labor demand evolves across occupations with different levels of AI exposure as generative AI adoption expands and AI capabilities advance.

Job postings. Our analysis relies on job posting data from Lightcast, a provider that compiles vacancies scraped from major online recruitment platforms. We use postings from November 2021 through April 2026 and focus on a subset of economies with sufficient coverage. The resulting sample contains approximately 700 million job postings across 110 economies. The data provide information on employers, occupations, industries, required skills, and remote-work arrangements. For a subset of postings, information on minimum years of experience required and wages is also available. Occupations are classified into 342

four-digit ISCO-08 occupations and industries into 20 two-digit NAICS sectors.[13] For this set of economies, the data also provides 600 million worker profiles and over 560,000 firms in the cross-section.

We arrive at our sample of 110 economies using a series of coverage and quality filters. First, we exclude economies with fewer than 1,000 postings in 2025. We then retain economies that satisfy at least one of two criteria measured using data from May 2025 through April 2026: (i) the ratio of active worker profiles to the 2025 labor force is at least 5 percent, or (ii) the total number of postings exceeds 25,000.[14] Appendix Figure A.1 displays the set of economies retained after applying these selection criteria. While we include a broad set of economies in our baseline sample, our results are robust to focusing on a smaller set of economies with better coverage.

We impose several additional restrictions on the postings data. We exclude postings from staffing agencies, postings with missing or unclassified industry information, and economy-industry-occupation-month cells containing fewer than three postings. We also exclude military and armed forces occupations.

In addition, the skill-level information contained in the Lightcast data allows us to observe the requirements employers specify in their vacancies. In Section 4, we use these data to construct a measure of AI adoption based on references to emerging AI skills in job postings. This measure can be tracked across economies, industries, occupations, firms, and time, providing a complementary perspective on how generative AI is reshaping labor demand.[15]

AI exposure. To measure occupational exposure to AI, we use the complementarity-adjusted AI Occupational Exposure (C-AIOE) index proposed by Pizzinelli et al. (2023). The measure is constructed by adjusting the AI Occupational Exposure (AIOE) index of Felten et al. (2021) for the degree of occupational complementarity with AI. Specifically,

[13] Information on industry classifications is available only for a subset of postings, reducing sample size for analyses where this classification is needed.

[14] Active profiles are those with locations updated after May 2025. This approach allows us to restrict attention to profiles with information updated recently.

[15] Appendix Table A.1 reports summary statistics. See Appendix Table A.2 for examples of job postings and listed skills.

the adjustment depends on an occupation-level complementarity score that aggregates six task characteristics: communication, responsibility, physical work, critical thinking, routine intensity, and skills requirements. The complementarity adjustment distinguishes occupations where AI is more likely to augment workers from those where it is more likely to substitute for them. As a result, occupations with high AI exposure but strong potential for human-AI complementarity are treated as facing lower displacement risk than equally exposed occupations whose tasks are more readily automated. The complementarity adjustment tends to reduce displacement risk for occupations involving greater responsibility, judgment, management, or other tasks where humans remain essential complements to AI.

AI capabilities. We also use information on frontier AI capabilities from METR, a nonprofit research organization that tracks when leading AI systems become capable of performing software tasks of increasing duration and complexity (METR, 2026). For example, at the time of ChatGPT's launch, AI was capable of completing tasks such as answering simple questions that would otherwise require well under a minute of human effort. AI models then became capable of finding a fact on the web (May 2024, 7 minute task), fixing bugs in small Python libraries (February 2025, 1 hour), and, more recently, training an adversarially robust image model (late 2025, 4 hours). METR records the first month in which an AI system is able to complete each benchmark task with 50 percent probability of success. We link these milestone dates to the monthly job posting data in order to study how labor-demand outcomes evolve as AI capabilities advanced over time.

3 Differential impact in more exposed occupations

This section examines how labor demand has evolved in more-exposed occupations compared to less-exposed occupations following the emergence of generative AI. We find that hiring declines significantly in more-exposed occupations relative to less-exposed occupations. Importantly, we show that this decline has largely stabilized even as frontier AI capabilities

continue to advance.

3.1 Empirical strategy

We begin by examining whether labor demand has evolved differently in occupations that are more exposed to AI following the launch of ChatGPT in November 2022. To do so, we estimate a dynamic difference-in-differences specification as in equation (1) that compares changes in job postings for highly exposed occupations relative to less exposed occupations before and after the introduction of generative AI.

$$Y_{eiot} = \sum_{k \neq -1} \beta_k \Big(\mathbf{1}_{o=High\ \ AI\ \ Exposure} \times \mathbf{1}_{t=AI\ \ Launch+k} \Big) + \alpha_{eit} + \alpha_{eio} + X_{eio} \sum_k \gamma_k \mathbf{1}_{t=k} + \varepsilon_{eiot}, \tag{1}$$

where e is economy, i is NAICS 2-digit industry, o is ISCO 4-digit occupation, and t is month. The unit of observation is an economy-industry-occupation-month cell. The outcome variable is new job postings in logs. We treat occupations with above median C-AIOE scores as in Pizzinelli et al. (2023) as highly exposed. This is a standard difference-in-differences specification: the timing of treatment is not staggered across units.

Importantly, our specification includes economy-industry-time fixed effects, economy-industry-occupation fixed effects, and event-time indicators interacted with each occupation's pre-period share of remote-work postings. The economy-industry-time fixed effects absorb shocks common to all occupations within an economy-industry-month, while the economy-industry-occupation fixed effects account for persistent differences across occupations within economy-industry pairs. The remote work interactions flexibly control for differential trends associated with varying scope for telework. As a result, identification comes from comparing occupations with different levels of AI exposure within the same economy, industry, and month, helping to isolate the effects of AI from other concurrent forces shaping labor demand, such as sector-specific hiring cycles or the post-pandemic expansion of remote work (Gimbel

et al., 2026; Lambert and Schindler, 2026). Standard errors are clustered at the occupation level.

The key identifying assumption is that, absent the launch of ChatGPT, labor demand trends would have evolved similarly across occupations with different levels of AI exposure after accounting for our controls. Because identification comes from within-economy-industry comparisons, a confounding factor would need to generate differential labor demand trends for highly exposed occupations across a broad set of economies at precisely the time generative AI emerged. Such shocks are considerably less plausible than economy- or industry-level explanations, which are absorbed by the fixed effects.

3.2 Estimates of the effect of AI on labor demand

Figure 1(a) presents our estimates of the dynamic differential impact of AI on employer demand in more-exposed occupations. Consistent with the identifying assumption, the figure shows no evidence of differential pre-trends in labor demand across occupations with different levels of AI exposure prior to the launch of ChatGPT. Following the launch of ChatGPT, labor demand declines in occupations with higher AI exposure relative to occupations with lower exposure. The estimates indicate that the differential effect was initially modest and became more negative through 2023 and 2024. By the later part of the sample, the estimated decline is nearly 10 percent.

Table 1(a) reports corresponding difference-in-differences estimates. Column (1) includes economy-month and economy-occupation fixed effects, while column (2) adds more granular economy-industry-month and economy-industry-occupation fixed effects. Column (3), our preferred specification, further controls for differential remote work trends across occupations. Across all specifications, the impact of AI on labor demand for highly exposed occupations relative to less exposed occupations is negative and statistically significant, and ranges from a 5 percent to 8 percent decrease. In our preferred specification, the pooled effect in the combined post-AI period is about 6 percent decrease in hiring for more-exposed

occupations relative to less-exposed occupations.

Together, the event-study and difference-in-differences results indicate that labor demand weakened in occupations with greater exposure to AI following the introduction of ChatGPT.[16] The dynamic estimates further show that the deterioration accumulated over the first two years after the launch of AI. Note that the estimates focus on the flow of job postings: the impact on the stock of employment will accumulate slowly over time and depend on other factors, including separations, turnover, and realized hiring.

3.3 Robustness

We next examine the robustness of our baseline findings. We consider alternative measures of labor demand, alternative definitions of AI exposure, different segments of the labor market, and alternative samples of economies. Across these exercises, the qualitative pattern remains similar to the baseline estimates.

Using different measures of labor demand. Appendix Table B.1 reports estimates using several alternative outcome variables. In place of log postings, we use the number of postings, the number of postings winsorized at the first and ninety-ninth percentiles, and occupation posting shares within economy-industry-time cells. The impact of AI remains negative in all specifications and statistically significant.

Using different measures of AI exposure. Our baseline specification classifies occupations using C-AIOE. Appendix Table B.2 shows that the results are similar when exposure is measured using the original AIOE index of Felten et al. (2021) or when exposure is measured as a continuous variable rather than being binned into above- or below-median. In both cases, the estimated coefficients remain negative and statistically significant.

Appendix Table B.3 categorizes occupations according to both exposure and complementarity. Relative to occupations with low exposure and low complementarity, occupations classified as high exposure and low complementarity experience the largest declines follow-

[16] We conduct a parallel analysis using wages as the outcome variable in the subset of postings that include wage information and find no evidence of a wage response. See Appendix Figure D.1 and D.2.

ing the launch of ChatGPT. Occupations that are high exposure and high complementarity also exhibit negative and statistically significant estimates, but the magnitude is smaller. By contrast, occupations that are low exposure and high complementarity display positive coefficients, although the estimate is statistically significant only in the full-sample specification. These results are consistent with the predictions of Pizzinelli et al. (2023), who argue that occupations with high AI exposure but greater complementarity with AI should be less adversely affected than equally exposed occupations whose tasks are more readily replaced by AI.

Seniority. For the subset of postings containing minimum experience requirements, we examine whether the effect differs for junior vs. senior hiring. Appendix Table B.4 reports estimates separately by years of required experience. We find that there is largely no statistically significant difference in labor demand for junior vs. senior roles.

Lightcast Tier 1 & 2 economies. Coverage varies across economies in the Lightcast data. To assess whether the baseline results are sensitive to differences in data availability, Appendix Table B.5 restricts the sample to a group of 30 economies with the best quality of data coverage.[17] The estimated coefficient remains negative and statistically significant, with a magnitude similar to that obtained in the full sample.

Results for Asian (APD) economies. Appendix Figure B.1 and Appendix Table B.6 repeat the analysis for APD economies. The event-study estimates display a pattern similar to that observed in the full sample, with labor demand declining in more exposed occupations after the launch of ChatGPT and subsequently stabilizing. The corresponding difference-in-differences estimates range between -6 to -7 percent and remain statistically significant across specifications.

[17] Tier 1 and Tier 2 economies are those for which Lightcast provides its most comprehensive labor market coverage, including detailed classifications of occupations, industries, and skills, richer employer- and worker-level information, salary data, and longer historical series. Focusing on these economies provides a useful robustness check against potential measurement error arising from lower coverage in other economies.

3.4 Event study estimates in AI capability space

The preceding analysis measures event time relative to the launch of ChatGPT. Since November 2022, however, AI capabilities have continued to improve substantially, feeding into anxieties that each new technological leap will affect more jobs. To examine how labor demand evolved alongside these advances, we link the job posting data to capability milestones compiled by METR (2026). Rather than indexing event time by calendar months, we group months according to the AI capability milestone prevailing at that point in time and estimate the same specification using these milestones as the event dimension.

Figure 1(b) reports the results. The estimated impact becomes more negative as frontier systems progress from the earliest capability milestones toward the milestone corresponding to one-hour tasks. Beyond that point, the effect remains negative but largely stabilizes.[18] The red dashed line pools observations after AI reaches the one-hour milestone and indicates an estimated effect of approximately 8 percent. These results suggest that the relative decline in labor demand stabilized despite subsequent advances in AI capabilities.

To our knowledge, this is the first evidence linking changes in labor demand to successive improvements in frontier AI capabilities. The finding that the relative decline in labor demand stabilized once AI models reached the one-hour milestone is somewhat surprising, as one might have expected continued capability gains to translate into progressively larger labor-market effects. In Section 4, we develop a new measure of AI adoption to better understand this pattern.

3.5 Heterogeneity in impact of AI across economies

The effect of AI exposure differs across groups of economies. To examine this heterogeneity, we estimate the baseline specification separately by income category and region, pooling observations on and after the one-hour capability milestone. Appendix Figure B.2 reports the resulting estimates.

[18] See Appendix Table B.7 for regressions separating the post-AI period along similar lines.

The estimates vary substantially across income groups. Advanced economies exhibit the largest negative effects, while the corresponding estimates are smaller for emerging market economies. For low-income economies, the effect is close to zero and not statistically significant. In terms of geographical heterogeneity, the results are broadly similar across regions (other than in Africa where the estimated impact is small and not statistically significant, consistent with the null result for low-income economies).[19] In addition to being of interest in their own right, these results show that are findings are not driven by individual countries.

4 Demand for emerging AI skills

The preceding analysis establishes that labor demand has fallen in more AI-exposed occupations but that this effect has stabilized despite continued advances in AI capabilities. This pattern raises an important question: if AI is becoming increasingly capable of performing complex tasks, why has the relative decline in labor demand not continued to deepen? Occupational exposure measures are not well-suited to answering this question: they represent ex-ante predictions of where AI is likely to have an impact, not ex-post measures of whether firms are actively adopting AI. To study this margin, we construct a novel measure of AI adoption based on references to emerging AI skills in job postings.[20] This measure enables us to track the adoption of AI across occupations and firms and to examine whether labor demand is being reshaped through changing skill requirements and firm-level adoption.

4.1 Identifying emerging AI skills

We identify emerging AI skills by examining changes in the prevalence of skill requirements over time as in Atalay et al. (2020). We begin by constructing a set of candidate AI skills consisting of 1,411 skills identified using keywords such as "AI," "Generative AI," and "Lan-

[19] Lightcast covers postings in local languages. The mapping from job descriptions to skills may not be equally effective for all languages. However, the broad similarity of results across regions with very different mixes of languages suggests that any associated measurement error is not an important driver of our findings.

[20] Appendix Table A.2 contains examples of job postings that reference these skills.

guage Model." For each skill, we then calculate the cumulative distribution of its share in job postings over the period from 2010 to 2026. We classify a skill as an emerging AI skill if its cumulative distribution is below 1 percent during 2010-2011 and exceeds 1 percent during 2023-2026. This procedure yields an initial set of 39 skills.

To validate and expand this classification, we compare the resulting skills against existing AI-skill taxonomies from Lightcast and Jaumotte et al. (2026). Repeating the same procedure using these external classifications identifies additional AI-related skills, yielding a final set of 51 emerging AI skills.

Using this classification, we identify postings that require at least one emerging AI skill and use their prevalence as a measure of AI adoption. Because the measure is constructed directly from job postings, it can be aggregated consistently across multiple dimensions of the labor market and observed at monthly frequency.

Appendix Figure C.1 shows the median share of postings at the economy level requiring an emerging AI skill increases markedly during the sample period, indicating rapid growth in employer demand for these skills. Appendix Figure C.2 shows that there is large variation in economy-level AI adoption.[21]

4.2 Demand for AI skills is increasing

Next, we examine how demand for emerging AI skills varies across occupations and over time. Figure 2 plots the share of postings requiring an emerging AI skill against the standardized occupational AI exposure measure. Each dot corresponds to a four-digit ISCO-08 occupation. The figure shows a positive relationship between AI exposure and the prevalence of AI-skill requirements. Occupations with higher exposure are more likely to post vacancies requiring emerging AI skills.

This validates our measure of exposure used in the difference-in-differences, as the occupations hypothesized to be more exposed to AI are indeed the ones that are adopting

[21] This variation across economies is particularly interesting given that our measure of AI adoption draws on data on online job postings that is by nature skewed towards formal-sector jobs across all economies.

AI. Furthermore, because these skill shifts are concentrated in highly exposed occupations, they strengthen the interpretation that the effects identified in our difference-in-differences framework are driven by AI rather than other labor market trends occurring at the same time.

Figure 3 examines the evolution of AI skill demand as AI capabilities improve over time. The share of postings containing at least one emerging AI skill rises substantially as the AI frontier models become increasingly capable, with the increase concentrated in occupations that are more exposed to AI.[22]

Taken together, these findings indicate that demand for emerging AI skills has increased rapidly since the introduction of generative AI and that this increase is concentrated in occupations with greater AI exposure. This signals that AI is not only reducing hiring in these occupations, but also changing the nature of the occupations themselves.

4.3 Evidence at the firm level: empirical strategy

The preceding analysis focuses on occupations. However, decisions about adopting new technologies are ultimately made by firms. We therefore turn to firm-level data to examine whether differences in AI adoption are associated with differences in labor-demand growth. Using our measure of emerging AI skills, we classify firms according to whether they have posted at least one vacancy requiring an emerging AI skill. We refer to such firms as AI-adopting firms.

$$Y_{eifo} = \beta_1 \mathbf{1}_{eif=AI\ \ Adopting\ \ Firm} + \beta_2 \mathbf{1}_{o=High\ \ AI\ \ Exposure} + \beta_3 \Big(\mathbf{1}_{eif=AI\ \ Adopting\ \ Firm} \times \mathbf{1}_{o=High\ \ AI\ \ Exposure} \Big) + \delta Firm_Size_{eif} + \alpha_{ei} + \varepsilon_{eifo} \quad (2)$$

where e is economy, i is NAICS 2-digit industry, f is firm identifier, and o is ISCO 4-digit

[22] The same pattern is visible in the more granular occupation-level scatter plots presented in Appendix Figures C.3(a)-C.3(d), which show that occupations with higher AI exposure account for an increasing share of AI skill postings over successive periods. Appendix Table B.8 confirms that AI adoption is indeed statistically significantly higher in more-exposed occupations.

occupation.

Our firm-level analysis uses data from the most recent year of the sample, spanning May 2025 through April 2026 with over 560,000 economy-industry-firms. The dependent variable is the change in log postings from the previous year at the economy-industry-firm-occupation level. The classification of whether firms adopt AI is also based on all postings over the previous year. We relate posting growth to indicators for AI adoption and occupational AI exposure while controlling for firm size, measured as lagged log postings, and including economy-industry fixed effects. Occupations are classified as highly exposed when their C-AIOE value exceeds the median. Standard errors are clustered at the firm level.

4.4 Larger firms are more likely to adopt AI

Before examining posting growth, we document the distribution of AI adoption across firms. Appendix Figure B.3 plots the relationship between firm size and emerging AI adoption. We document that larger firms are more likely to post vacancies requiring emerging AI skills, while many smaller firms do not appear to do so.

Even within large firms, under 20 percent of firms adopt AI according to our measure. Note that this should be interpreted as a lower bound for the fraction of firms that adopt AI, as this captures firms that want new employees to have AI skills. Firms could also adopt AI in ways that simply reduces new hiring needs, for example, by training existing workers to leverage AI or substituting away from labor to AI.

4.5 AI adopting firms have higher growth

We next examine whether AI adoption is associated with broader differences in hiring activity. Table 1(b) reports firm-level regression estimates. Across specifications, the coefficient on the AI-adoption indicator is positive and statistically significant. In the specification without the interaction term, AI-adopting firms exhibit higher posting growth than firms that do not adopt AI.

Table 1(b) also confirms that posting growth is lower in occupations classified as highly exposed to AI. When the interaction between AI adoption and high AI exposure is included, the interaction term is negative and statistically significant, although the coefficient on AI adoption remains positive and statistically significant.

Figure 4 presents the corresponding marginal effects derived from the specification with the interaction term. The figure shows that firms classified as AI adopters exhibit higher posting growth than non-adopting firms in both low-exposure and high-exposure occupations. These results focus on growth in postings. It is also important to note that hiring at firms that adopt AI per our measure tends to be somewhat more concentrated in high-exposure occupations.

These findings are robust to alternative specifications. In Appendix Table B.9, we consider alternative samples of economies with better data coverage as well as alternative measures of AI exposure. Across all specifications, AI adopting firms experience higher posting growth than their non-adopting peers, and this pattern is true even in the more exposed occupations.

Taken together, the firm-level evidence indicates that firms posting vacancies requiring emerging AI skills tend to experience faster growth in job postings. Combined with the rising demand for AI skills documented above, these findings suggest that employer demand is increasingly incorporating emerging AI capabilities while continuing to evolve across occupations and firms.

AI's overall impact thus reflects two forces at once: less hiring in occupations where AI is more useful, and new opportunities for the workers who can put it to use.[23] The evidence to date suggests that reduced hiring remains more important, explaining why demand has fallen

[23]The fact that most firms have not yet adopted AI also points to the importance of barriers to incorporating this new technology into existing workflows and processes. For example, even if AI can perform tasks previously handled by individual workers, it may not be obvious how to use the technology to make decisions currently made by groups of people. Newer models may also be more expensive to use with larger energy footprints. Moreover, firms may be experimenting with AI before making decisions about hiring needs. Indeed, the full impact of the introduction of electric power took decades to emerge; many firms did not adopt the technology and continued to organize factory layouts around mechanical power generated by water or steam. (David, 1990)

in more exposed occupations. But offsetting opportunities have begun to emerge already for workers with the right skills.

5 Conclusion

The rapid advance of generative AI has intensified interest in how new technologies may affect labor demand. While a growing literature has begun to examine these questions, the existing evidence has not reached consensus on the effect and largely focuses on the United States and select European economies. Moreover, advances in AI capability have received little attention. In this paper, we use nearly 700 million online job postings from 110 economies to examine how labor demand has evolved as AI has improved.

Using variation in occupational AI exposure and the November 2022 launch of ChatGPT, we find that labor demand declined in more AI-exposed occupations relative to less exposed occupations. Event-study estimates indicate that the differential decline emerged after the launch of ChatGPT, became larger through 2023 and 2024, and subsequently stabilized.

To better understand why the impact on labor demand appears to have stabilized despite continued advances in AI capabilities, we develop a novel measure of AI adoption based on references to emerging AI skills in job postings. We identify a set of emerging AI skills and show that demand for these skills has risen rapidly since 2023. Demand for emerging AI skills is concentrated in occupations with greater AI exposure and has become increasingly important over time. At the firm level, we find that firms posting vacancies requiring emerging AI skills tend to exhibit stronger posting growth than firms that do not do so.

Taken together, the evidence suggests that generative AI is associated with both reductions in labor demand in more exposed occupations and growing demand for workers possessing emerging AI skills. The findings point to substantial heterogeneity in how labor demand evolves across occupations and firms as AI adoption expands. More broadly, they indicate that understanding the labor-market consequences of AI requires not only measur-

ing occupational exposure, but also tracking the extent to which firms incorporate AI-related capabilities into production and hiring requirements.

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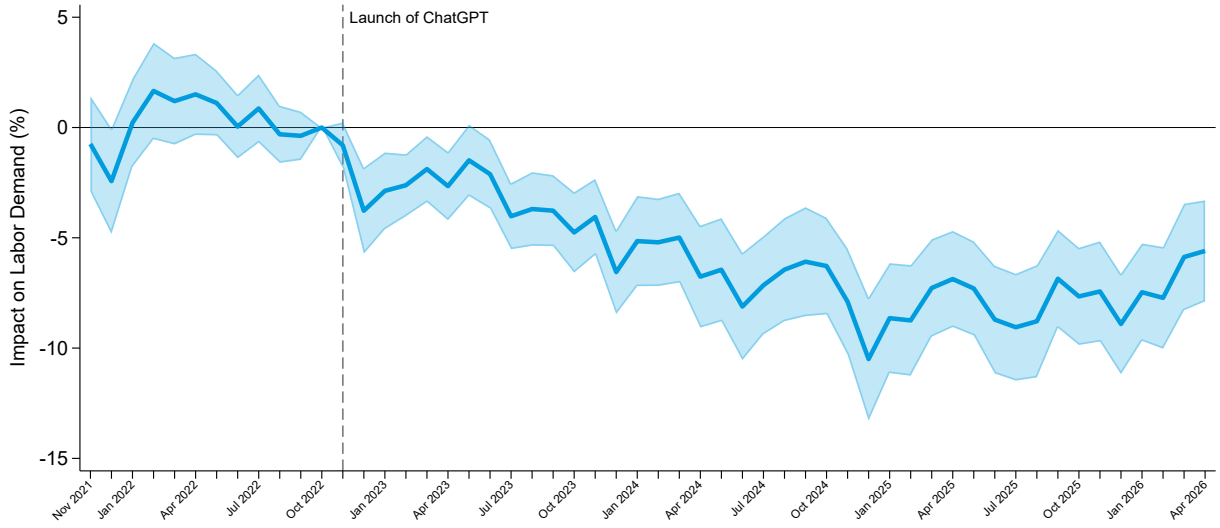
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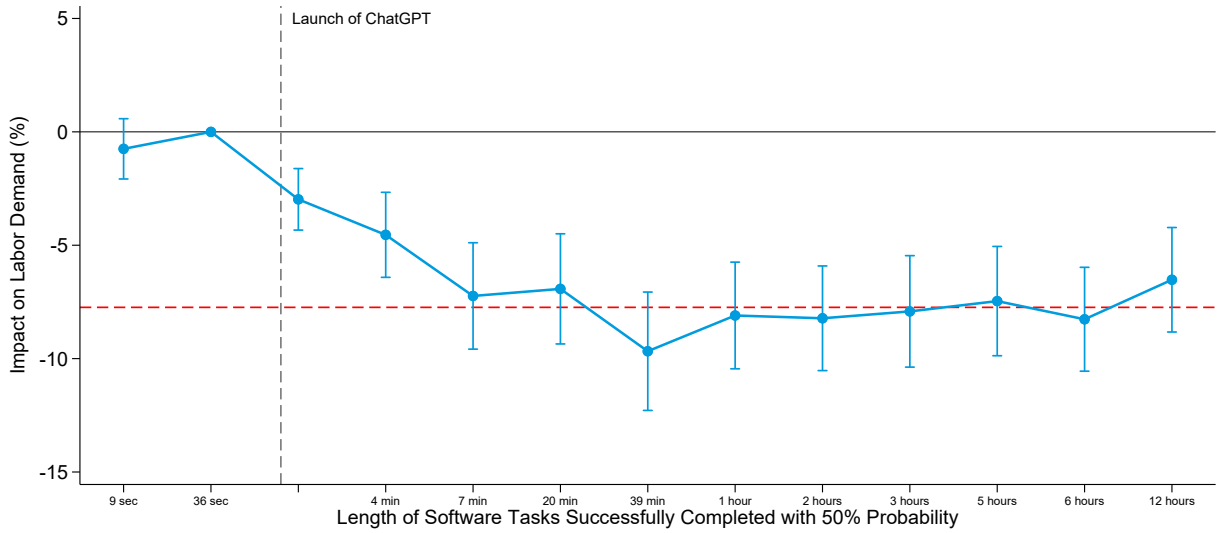
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Figure 1: Differential impact on demand in more exposed occupations as AI has improved



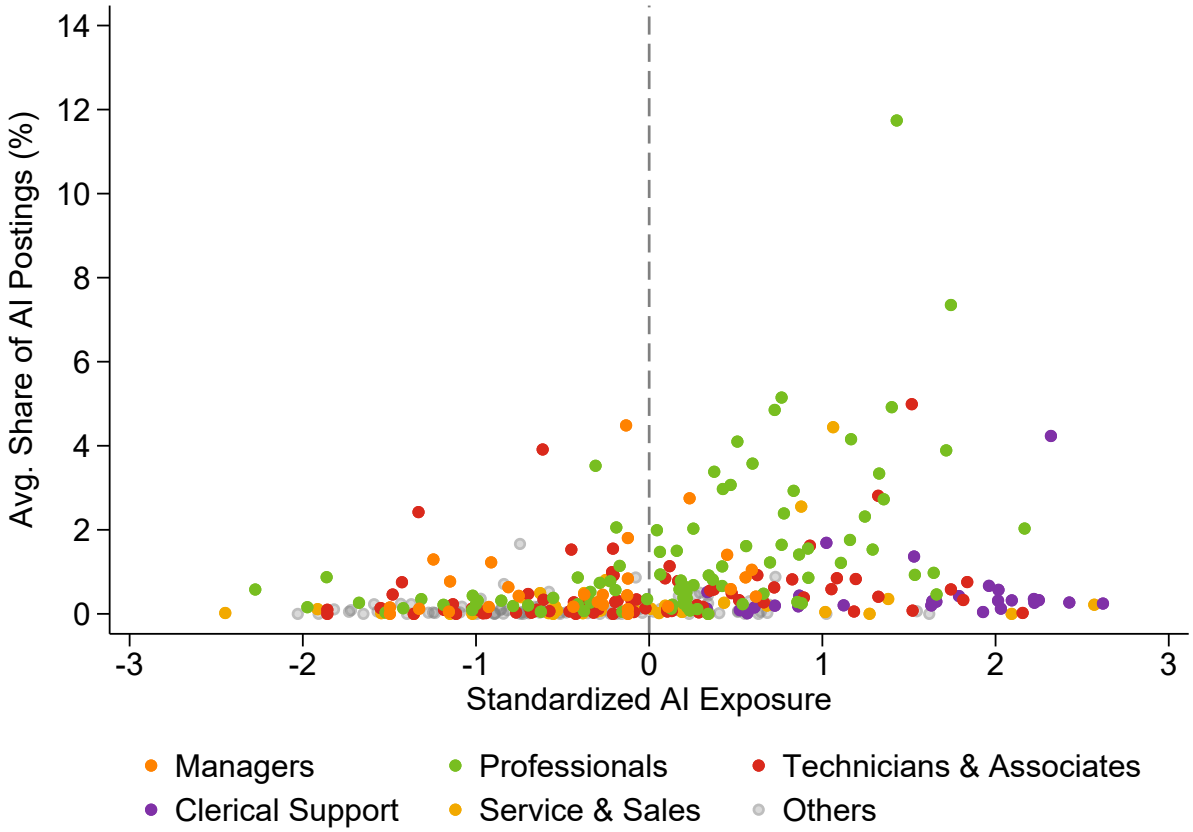
(a) Impact on labor demand over time



(b) Impact on labor demand and AI capability

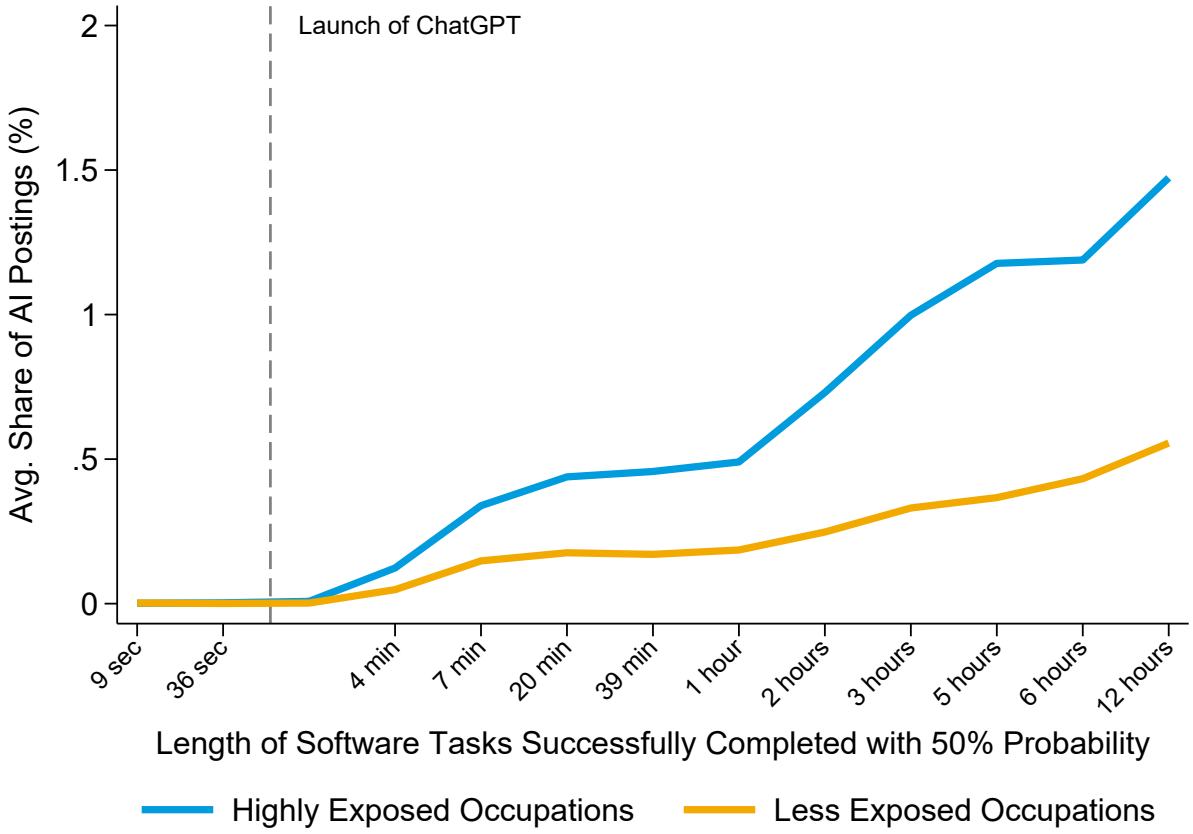
Notes: The sample comprises all economies. The figure shows event study estimates from equation (1). The red dashed line indicates the impact on labor demand pooling observations after AI reached the 1-hour mark. Standard errors are clustered at the occupation level. Shaded bands and whiskers show 95 percent confidence intervals.

Figure 2: Demand for emerging AI skills across occupations



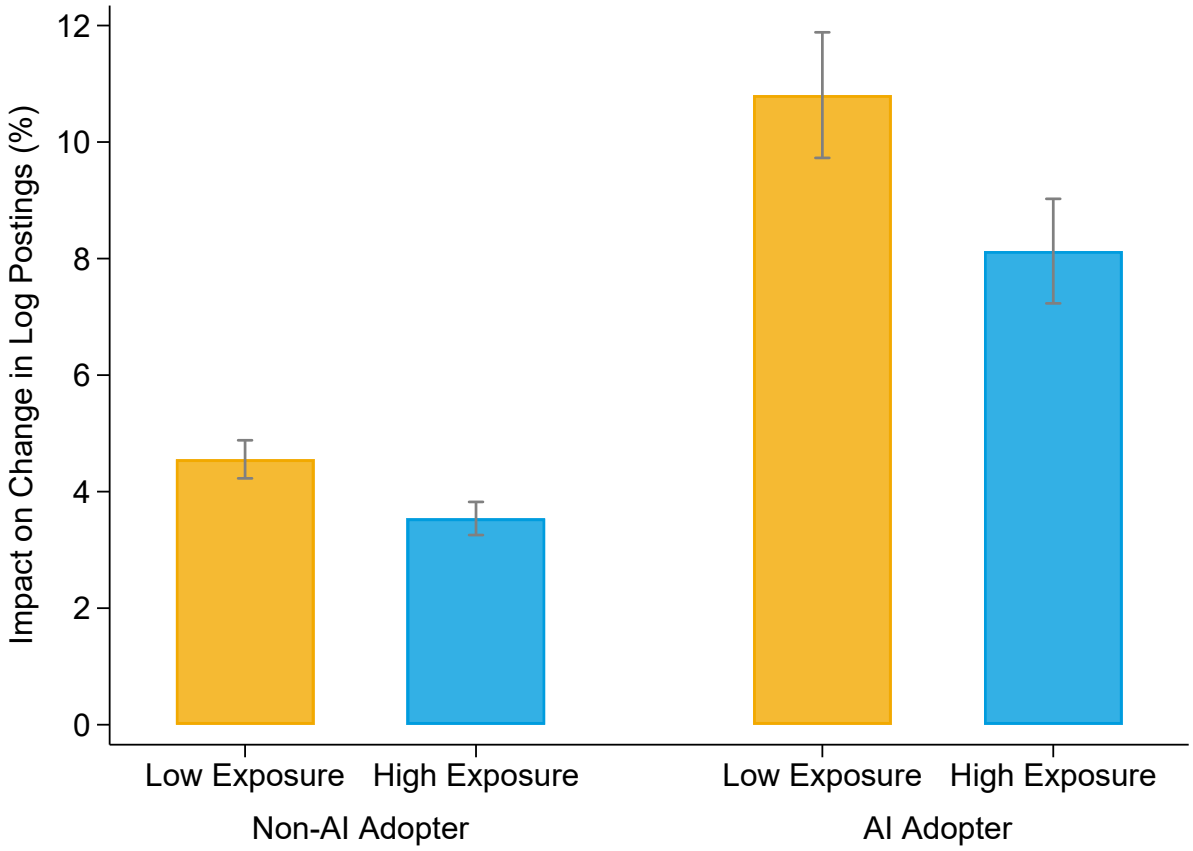
Notes: The sample comprises all economies. Share of AI postings is calculated as the share of postings that feature an emerging AI skill from November 2025-April 2026. Dots represent averages across all postings from November 2025-April 2026 in ISCO-08 4-digit occupations. Standardized AI Exposure Index is the z-score of the C-AIOE measure for each ISCO-08 4-digit occupation from Pizzinelli et al. (2023).

Figure 3: Emerging AI skills are increasingly important in more exposed occupations



Notes: The sample comprises all economies. Share of AI postings is calculated as the share of postings that feature an emerging AI skill, averaged across economy-industry-occupation cells.

Figure 4: Postings are growing faster at AI adopting firms



Notes: The sample comprises all economies. Marginal effects for each subgroup are calculated using regression estimates from Column (3) in Table 1(b).

Table 1: Differential impact of AI on labor demand in more exposed occupations

	(1)	(2)	(3)
High AI Exposure × Post AI Launch	-5.84***	-7.59***	-6.01***
	(0.92)	(1.18)	(1.01)
Economy × Time FE	Y		
Economy × Occupation FE	Y		
Economy × Industry × Time FE		Y	Y
Economy × Industry × Occupation FE		Y	Y
Remote Share Control	N	N	Y
Num. Economies	110	110	110
Num. Occupations	340	340	340
Economy × Industry × Occupation × Months	3,481,627	3,435,741	3,435,741
Total Num. Postings (mil)	203.1	202.9	202.9

(a) Difference-in-differences

	(1)	(2)	(3)
High AI Exposure	-1.33***	-1.26***	-1.02***
	(0.12)	(0.12)	(0.13)
Standardized Firm Size	-6.79***	-6.85***	-6.87***
	(0.16)	(0.16)	(0.16)
AI Adopt	4.95***	5.06***	6.25***
	(0.51)	(0.51)	(0.61)
AI Adopt × High AI Exposure			-1.66***
			(0.39)
Economy FE	Y		
Industry FE	Y		
Economy × Industry FE		Y	Y
Num. Firms	566,066	565,992	565,992
Num. Occupations	340	340	340
Economy × Industry × Firm × Occupations	2,826,298	2,826,224	2,826,224

(b) Firm regressions

Notes: The sample comprises all economies. Panel (a) reports the regression estimates from a version of equation (1) where the High AI Exposure dummy is interacted with a dummy for the post-treatment period. Panel (b) reports the regression estimates from equation (2). Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level in Panel (a) and at the economy-firm level in Panel (b), are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Online Appendices

A Lightcast data

A.1 Sample selection

Figure A.1: Set of economies after sample selection



Notes: Sample selection relies on postings and active profiles between May 2025 and April 2026.

Table A.1: Summary statistics

	Mean	SD	25th	Median	75th
Num. Postings	58.29	631.35	4.00	8.00	22.00
Pre-Period Remote Share (%)	7.34	11.12	0.00	2.78	10.50
C-AIOE	4.63	0.51	4.28	4.61	4.94
Share of AI Postings (bp)	36.81	316.72	0.00	0.00	0.00

Notes: The sample comprises all economies.

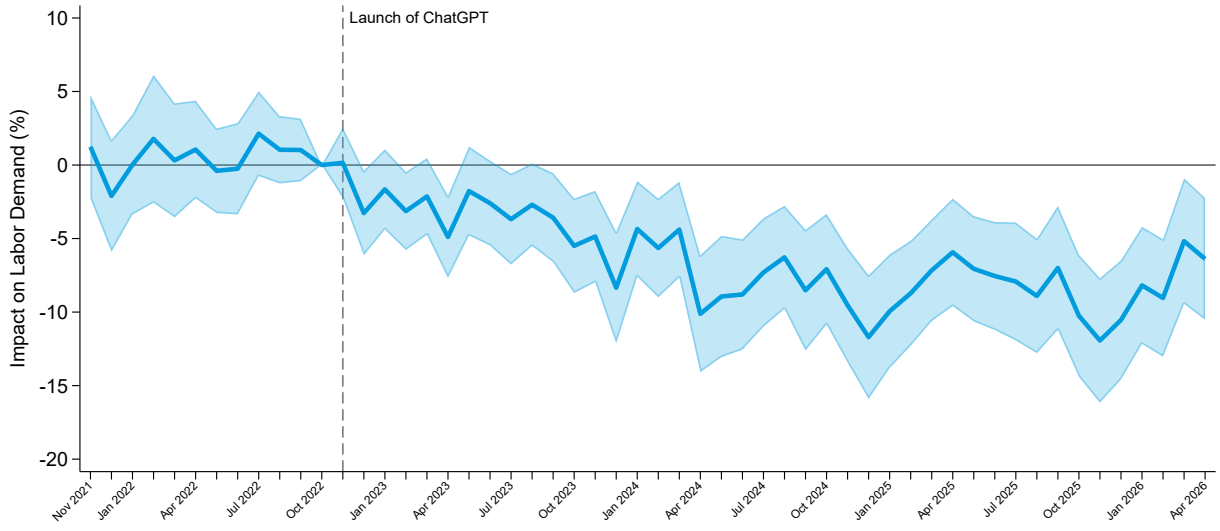
A.2 Example job postings

Table A.2: Excerpts from job postings in the Lightcast data

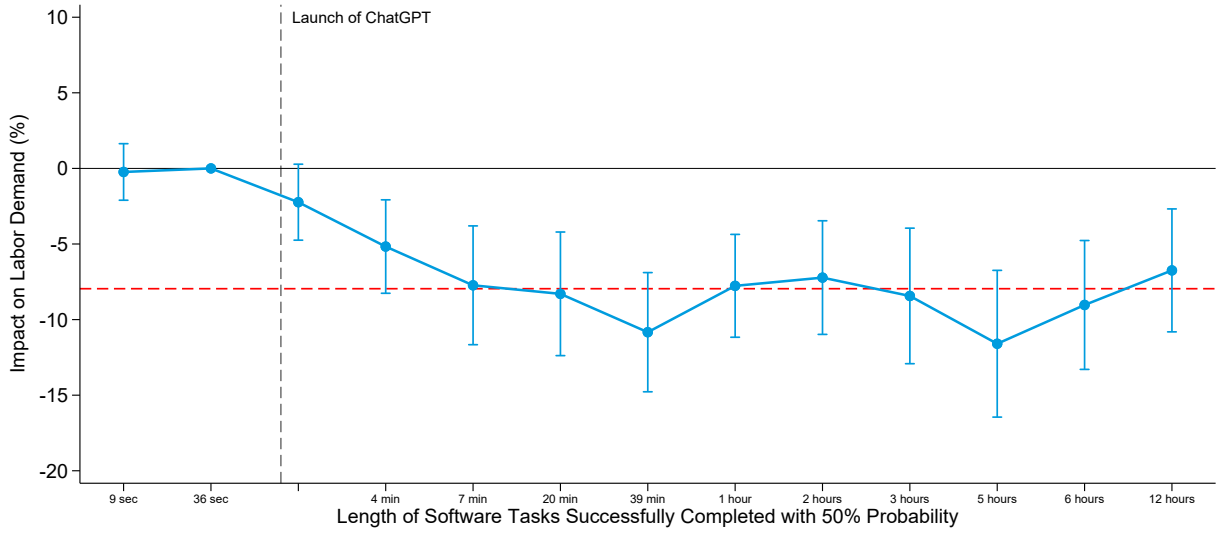
Country / Date	Company / Title	Skills	Summary of Role	Excerpt
Sri Lanka 27 Feb 2026	**IFS** Enterprise Risk Management Interns	AI Agents; Enterprise Risk Mgmt; Internal Auditing; ISO 31000; Risk Register	ERM and Internal Audit Lead role.	"Conduct external risk assessment with the use of AI and enhance the AI Agent in the process with the tech teams."
China 15 Apr 2026	**HSBC** Associate Directors	Generative AI; AI Agents; Java; Python; Google Cloud Platform	Senior technology SME in Guangzhou leading large delivery teams.	"Hands on experience of building AI agents"; "Using AI Coding toolings intensively in daily coding work"
Malaysia 19 Feb 2026	**Standard Chartered Bank** Model Validation Analysts	Retrieval Augmented Generation; Prompt Engineering; Model Risk Mgmt; Explainable AI; Python/R/SAS	Model validation manager independently validating models used in financial crime, operational, and AI risk.	"LLM architecture, prompt engineering, finetuning, Retrieval Augmented Generation (RAG), evaluation framework and benchmark evaluation of model performance including accuracy, stability, fairness"
Mexico 26 Mar 2026	**Blend Agency Ltd** Artificial Intelligence Engineers	AI Agents; Agentic Systems; RAG; LangChain; Responsible AI	Firm's first AI Engineer Lead, owning architecture and production deployment of generative-AI systems.	"Architect and lead the implementation of production-grade AI solutions leveraging LLMs, transformer-based models, RAG pipelines, and agentic systems."
Ireland 8 Apr 2026	**Citigroup** Lead Application Analysts	LangChain; Prompt Engineering; RAG; Vector Database; MLOps	GenAI applications technology lead combining generative-AI platform builds with fraud-detection model validation.	"Strong programming skills in Python, and familiarity with libraries like Transformers, LangChain, LlamaIndex, PyTorch, or TensorFlow"

B Additional figures & tables

Figure B.1: Response is similar in APD economies



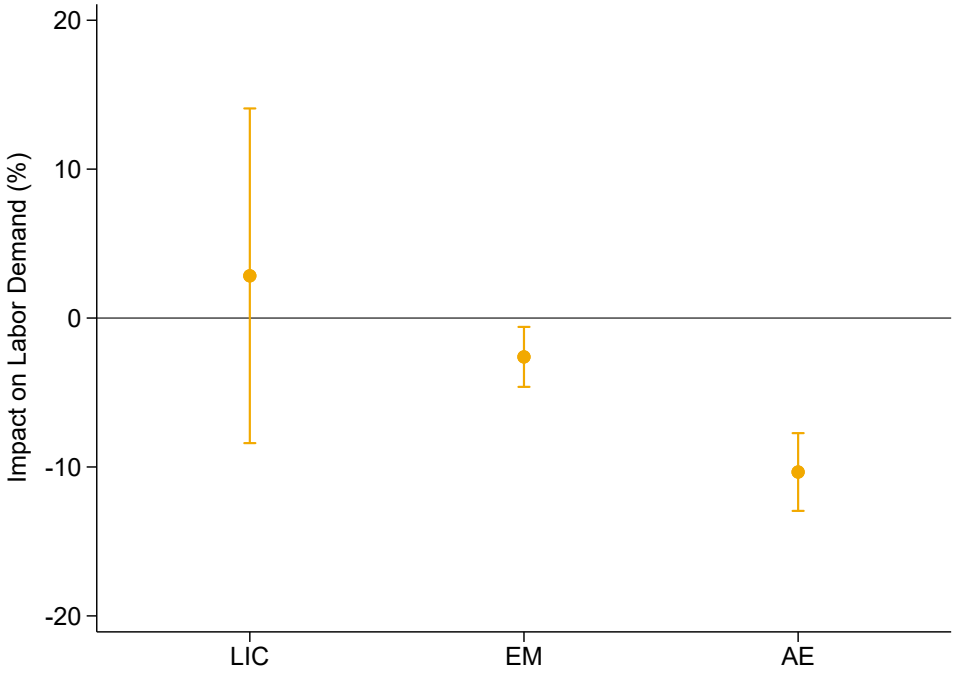
(a) Event study estimates



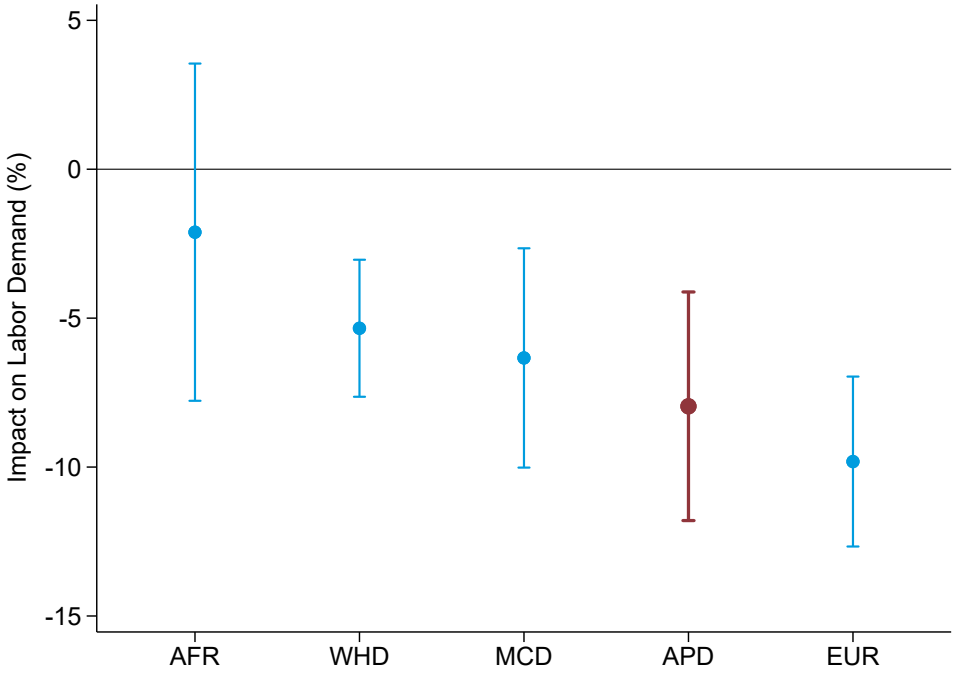
(b) Event study estimates and AI capability

Notes: The sample comprises APD economies. The figure shows event study estimates from equation (1). The red dashed line indicates the impact on labor demand pooling observations after AI reached the 1-hour mark. Standard errors are clustered at the occupation level. Shaded bands and whiskers show 95 percent confidence intervals.

Figure B.2: Estimates by income and region



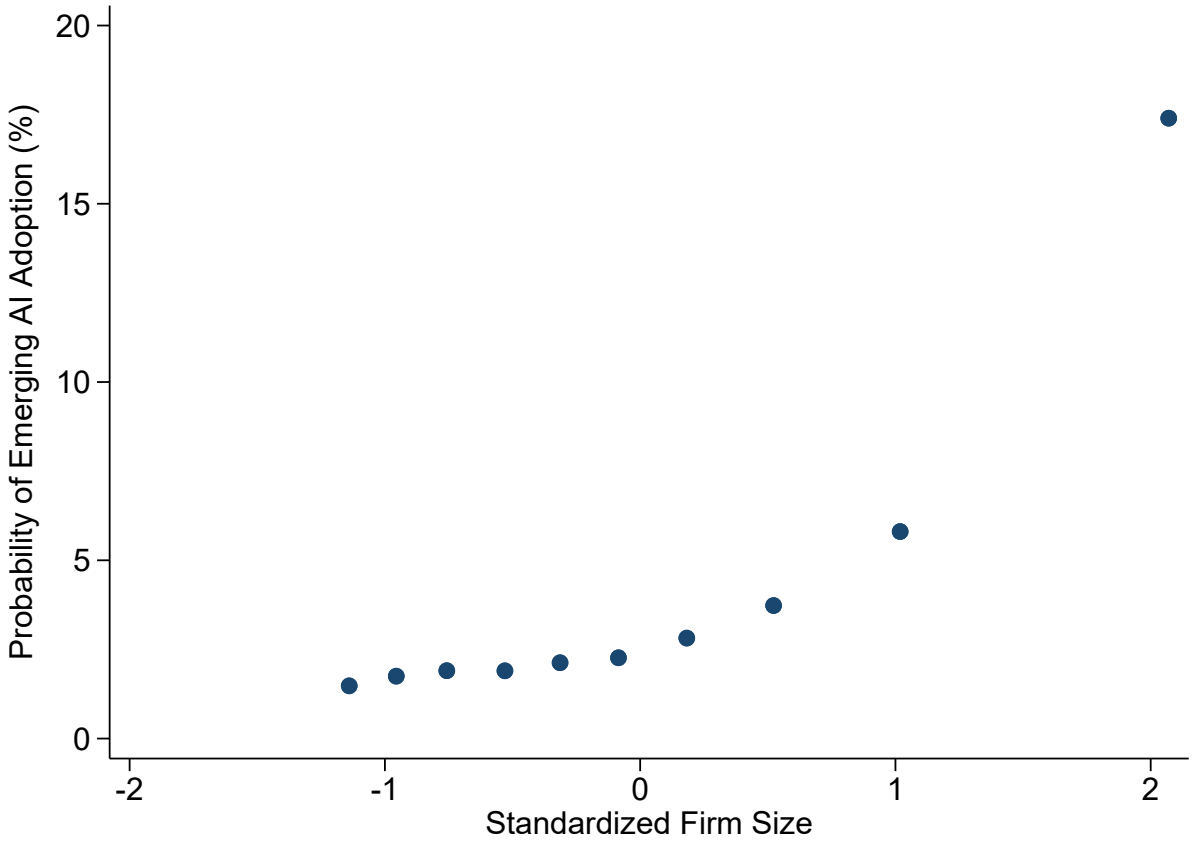
(a) By income group



(b) By region

Notes: The sample comprises all economies. The figure shows income- and region-specific estimates from a regression that pools observations after AI reached the 1-hour mark. Standard errors are clustered at the occupation level. Whiskers show 95 percent confidence intervals.

Figure B.3: AI adoption and firm size



Notes: The sample comprises all economies. Firm size is calculated as log postings from May 2024-April 2025, winsorized at the 1st and 99th percentiles, and standardized to have mean 0 and standard deviation 1. Standardized firm size and emerging AI adoption probabilities are adjusted for economy and industry fixed effects.

Table B.1: Using different measures of labor demand

	(1) Num. Postings	(2) Num. Postings (winsor)	(3) Share (EIT)
High AI Exposure × Post AI Launch	-8.34***	-3.18***	-0.04**
	(2.09)	(0.67)	(0.02)
Economy × Industry × Time FE	Y	Y	Y
Economy × Industry × Occupation FE	Y	Y	Y
Remote Share Control	Y	Y	Y
Num. Economies	110	110	110
Num. Occupations	340	340	340
Economy × Industry × Occupation × Months	3,435,741	3,435,741	3,435,741
Total Num. Postings (mil)	202.9	202.9	202.9

Note: The sample comprises all economies. Outcomes in model 2 are winsorized at the 1st and 99th percentiles. The coefficients in model 3 are multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Table B.2: Using different measures of AI exposure

	(1) AIOE	(2) Cts. C-AIOE	(3) Within 1 SD
High AI Exposure × Post AI Launch	-6.05***		
	(0.83)		
AI Exposure × Post AI Launch		-6.73***	
		(0.95)	
High AI Exposure × Post AI Launch			-4.15***
			(1.24)
Economy × Industry × Time FE	Y	Y	Y
Economy × Industry × Occupation FE	Y	Y	Y
Remote Share Control	Y	Y	Y
Num. Economies	110	110	109
Num. Occupations	340	340	236
Economy × Industry × Occupation × Months	3,435,741	3,435,741	2,292,063
Total Num. Postings (mil)	202.9	202.9	139.0

Notes: The sample comprises all economies. Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. The measure of high AI exposure is above-median C-AIOE, above-median AIOE, and C-AIOE above median and for the sample of occupations within 1 standard deviation of the median for models (1), (2), and (3), respectively. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Table B.3: Using different measures of AI exposure: high/low exposure + complementarity

	(1) APD	(2) All
High Exposure-High Complement. × Post AI Launch	-7.33***	-4.69***
	(1.73)	(1.21)
High Exposure-Low Complement. × Post AI Launch	-11.94***	-9.51***
	(1.98)	(1.43)
Low Exposure-High Complement. × Post AI Launch	1.08	2.46**
	(1.51)	(1.04)
Economy × Industry × Time FE	Y	Y
Economy × Industry × Occupation FE	Y	Y
Remote Share Control	Y	Y
Num. Economies	19	110
Num. Occupations	340	340
Economy × Industry × Occupation × Months	815,458	3,435,741
Total Num. Postings (mil)	21.6	202.9

Notes: Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Table B.4: Junior vs. senior postings

	(1) Jr ≤ 1	(2) Jr ≤ 3	(3) Jr ≤ 5	(4) Jr ≤ 7
Senior: High AI Exposure × Post AI Launch	-8.906***	-7.582***	-7.567***	-6.457***
	(1.443)	(1.522)	(2.151)	(2.170)
Junior - Senior Diff	2.863*	-0.128	-0.284	-1.307
	(1.641)	(1.228)	(1.748)	(1.824)
Economy × Industry × Time FE	Y	Y	Y	Y
Economy × Industry × Occupation FE	Y	Y	Y	Y
Remote Share Control	Y	Y	Y	Y
Num. Economies	103	106	103	103
Num. Occupations	340	340	340	340
Economy × Industry × Occupation × Months	1,590,317	1,635,180	1,442,379	1,388,236
Total Num. Postings (mil)	76.9	76.9	76.9	76.9

Notes: Junior-Senior Diff is the additional Post AI Launch effect for junior postings relative to senior postings among high-AI exposure occupations. The sample comprises all economies and postings with non-missing minimum years of experience. Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Table B.5: Using Lightcast Tier 1 & 2 economies

	(1)
High AI Exposure × Post AI Launch	-6.63***
	(1.08)
Economy × Industry × Time FE	Y
Economy × Industry × Occupation FE	Y
Remote Share Control	Y
Num. Economies	30
Num. Occupations	340
Economy × Industry × Occupation × Months	2,446,008
Total Num. Postings (mil)	187.9

Notes: The sample comprises economies classified by Lightcast as Tier 1 or Tier 2 based on the quality of coverage. Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Table B.6: Repeat Table 1 for APD

	(1)	(2)	(3)
High AI Exposure × Post AI Launch	-6.04***	-6.97***	-6.74***
	(1.46)	(1.71)	(1.68)
Economy × Time FE	Y		
Economy × Occupation FE	Y		
Economy × Industry × Time FE		Y	Y
Economy × Industry × Occupation FE		Y	Y
Remote Share Control	N	N	Y
Num. Economies	19	19	19
Num. Occupations	340	340	340
Economy × Industry × Occupation × Months	824,082	815,458	815,458
Total Num. Postings (mil)	21.6	21.6	21.6

(a) Difference-in-differences

	(1)	(2)	(3)
High AI Exposure	-2.51***	-2.32***	-2.31***
	(0.39)	(0.39)	(0.44)
Standardized Firm Size	-8.35***	-8.39***	-8.39***
	(0.61)	(0.59)	(0.59)
AI Adopt	7.70***	7.70***	7.73***
	(1.29)	(1.26)	(1.61)
AI Adopt × High AI Exposure			-0.04
			(1.05)
Economy FE	Y		
Industry FE	Y		
Economy × Industry FE		Y	Y
Num. Firms	69,472	69,467	69,467
Num. Occupations	340	340	340
Economy × Industry × Firm × Occupations	387,787	387,782	387,782

(b) Firm regressions

Notes: The sample comprises APD economies. Panel (a) reports the regression estimates from a version of equation (1) where the High AI Exposure dummy is interacted with a dummy for the post-treatment period. Panel (b) reports the regression estimates from equation (2). Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level in Panel (a) and at the economy-firm level in Panel (b), are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Table B.7: Differential impact of AI on labor demand before/after 1-hour capability milestone

	(1)	(2)	(3)
High AI Exposure × AI Launch to Pre 1-hour	-5.39***	-6.81***	-5.11***
	(0.95)	(1.20)	(1.01)
High AI Exposure × Post 1-hour	-6.68***	-9.11***	-7.74***
	(0.97)	(1.25)	(1.14)
Economy × Time FE	Y		
Economy × Occupation FE	Y		
Economy × Industry × Time FE		Y	Y
Economy × Industry × Occupation FE		Y	Y
Remote Share Control	N	N	Y
Num. Economies	110	110	110
Num. Occupations	340	340	340
Economy × Industry × Occupation × Months	3,481,627	3,435,741	3,435,741
Total Num. Postings (mil)	203.1	202.9	202.9

Notes: The sample comprises all economies. Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

Table B.8: AI adoption is more prevalent in more-exposed occupations

	(1) ET + EO FEs	(2) EIT + EIO FEs	(3) Tier 1/2
High AI Exposure × Post AI Launch	0.26**	0.22**	0.23**
	(0.10)	(0.10)	(0.09)
Economy × Time FE	Y		
Economy × Occupation FE	Y		
Economy × Industry × Time FE		Y	Y
Economy × Industry × Occupation FE		Y	Y
Remote Share Control	N	N	Y
Num. Economies	110	110	30
Num. Occupations	340	340	340
Economy × Industry × Occupation × Months	3,481,627	3,435,741	2,446,008
Total Num. Postings (mil)	203.1	202.9	187.9

Notes: The table reports the regression estimates from a version of equation (1) where the High AI Exposure dummy is interacted with a dummy for the post-treatment period, and the outcome variable is share of postings that ask for at least one emerging AI skill. In models (1) and (2), the sample comprises all economies. In model (3), the sample comprises economies classified by Lightcast as Tier 1 or Tier 2 based on the quality of coverage. Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

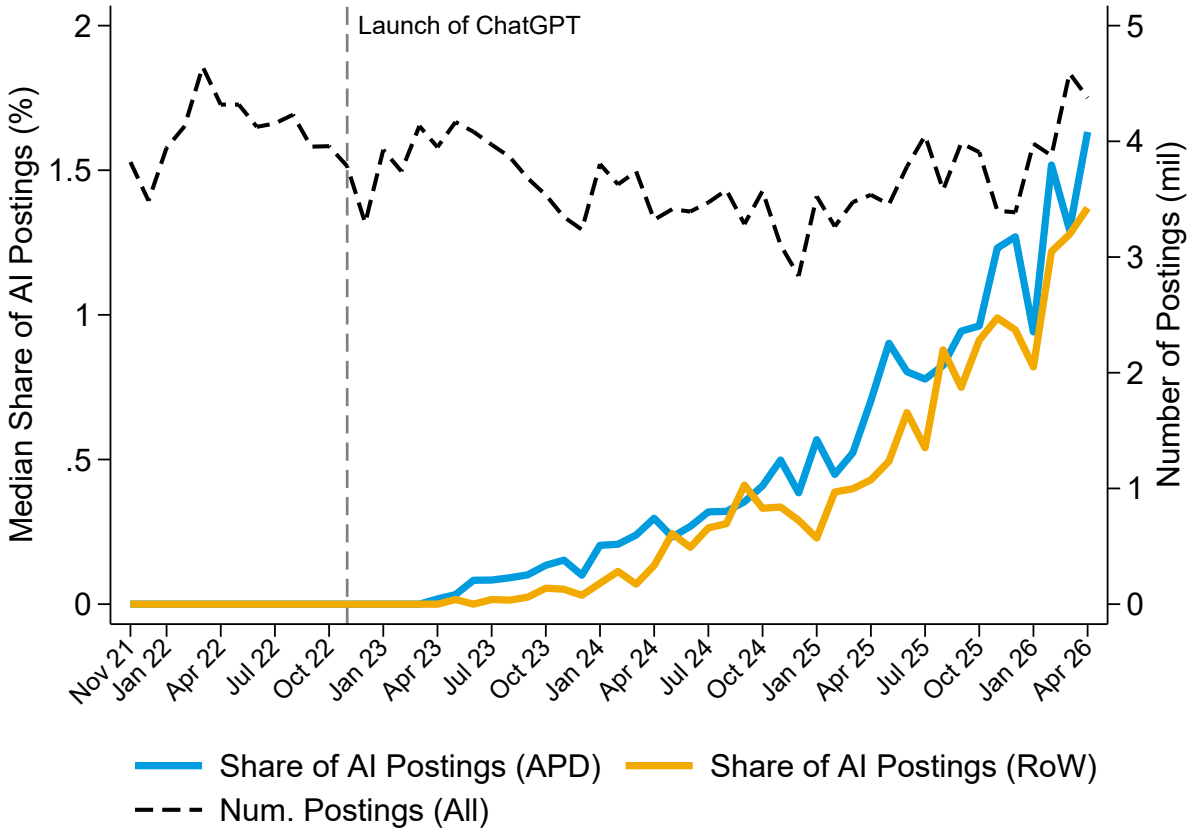
Table B.9: Alternative specifications of Table 1, Panel (b)

	(1) Tier 1/2	(2) AIOE	(3) Within 1SD
High AI Exposure	-1.16***	1.08***	0.13
	(0.14)	(0.16)	(0.16)
Standardized Firm Size	-6.74***	-6.97***	-7.10***
	(0.17)	(0.16)	(0.17)
AI Adopt	6.12***	9.22***	6.51***
	(0.64)	(0.75)	(0.64)
AI Adopt × High AI Exposure	-1.53***	-4.96***	-1.74***
	(0.41)	(0.59)	(0.42)
Economy × Industry FE	Y	Y	Y
Num. Firms	512,041	565,992	470,791
Num. Occupations	340	340	236
Economy × Industry × Firm × Occupations	2,538,195	2,826,224	1,891,398

Notes: The table reports the regression estimates from equation (2). In model (1) the sample comprises economies classified by Lightcast as Tier 1 or Tier 2 based on the quality of coverage. In models (2) and (3) the sample comprises all economies. The measure of high AI exposure is above-median C-AIOE, above-median AIOE, and C-AIOE above median and for the sample of occupations within 1 standard deviation of the median for models (1), (2), and (3), respectively. Coefficients have been multiplied by 100. Standard errors, clustered at the occupation level, are reported in parentheses. Statistical significance is denoted by *, **, and *** at the 10 percent, 5 percent, and 1 percent levels, respectively.

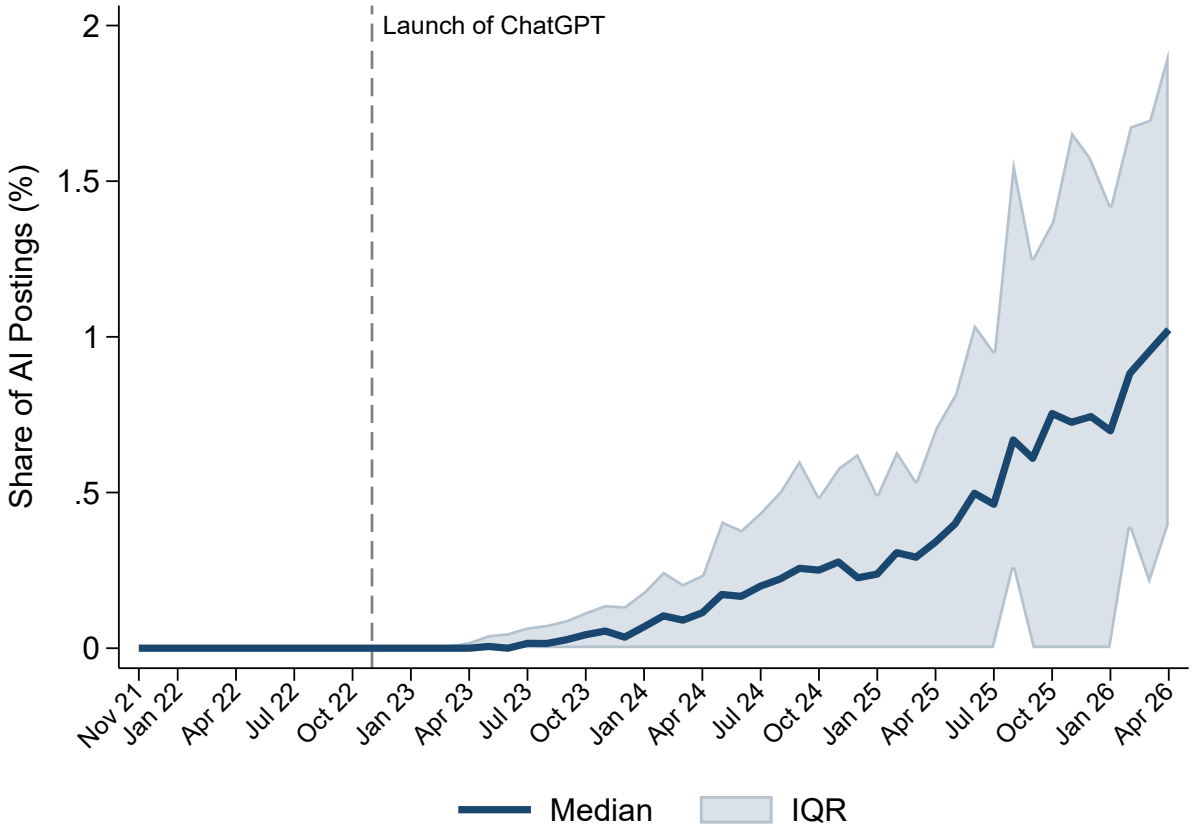
C Emerging AI skills

Figure C.1: Tracking AI with global real-time labor market data



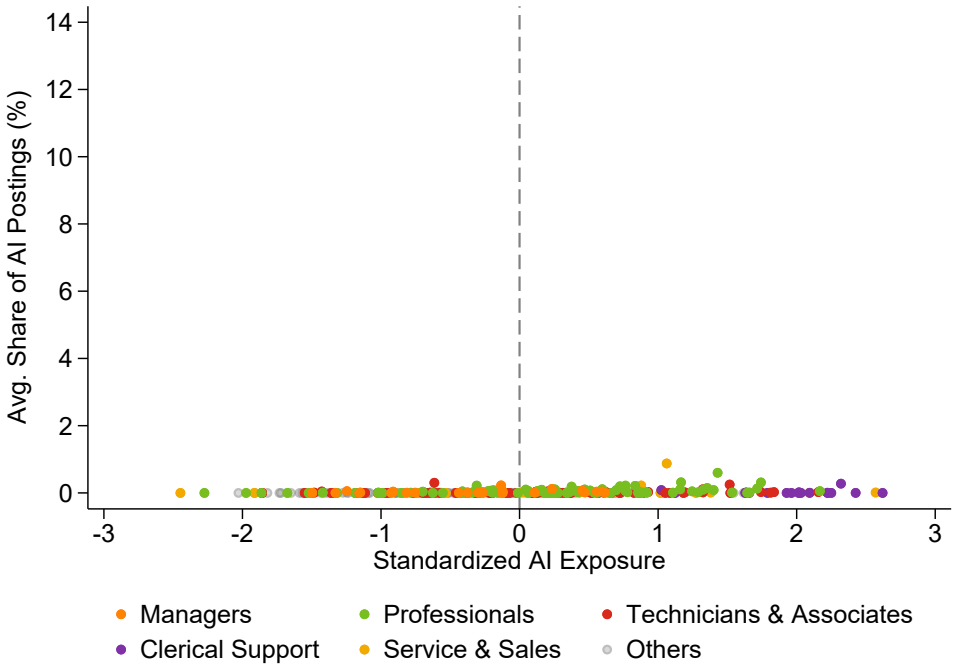
Notes: Share of AI postings is calculated as the median economy-level share of postings requiring at least one emerging AI skill in a given month.

Figure C.2: Variation in AI adoption across economies

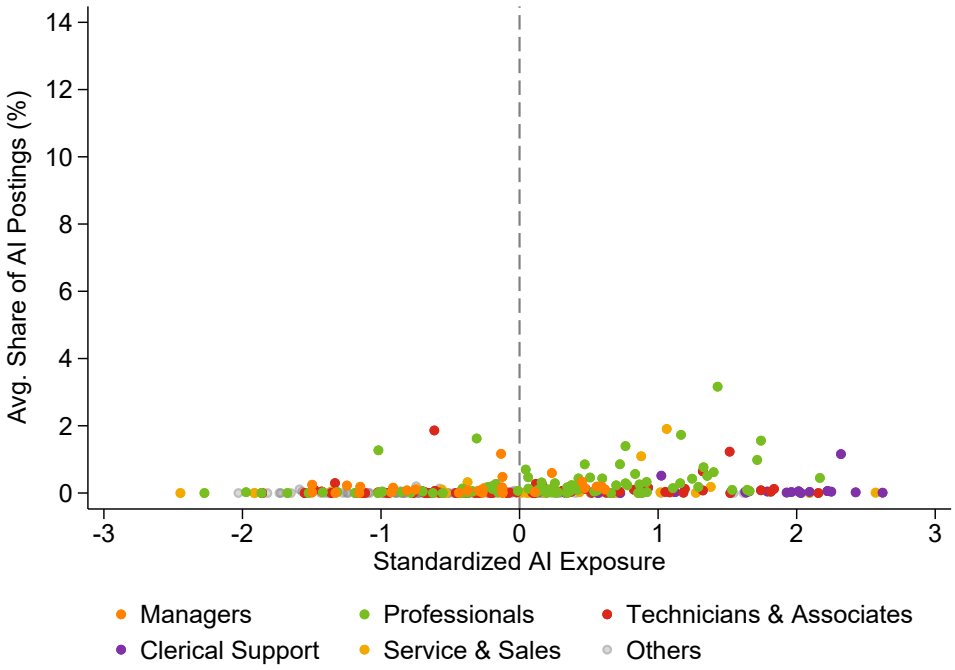


Notes: The sample comprises all economies. Share of AI postings is calculated as the median economy-level share of postings requiring at least one emerging AI skill in a given month. Shaded bands show the interquartile range.

Figure C.3: Emerging AI skills over time



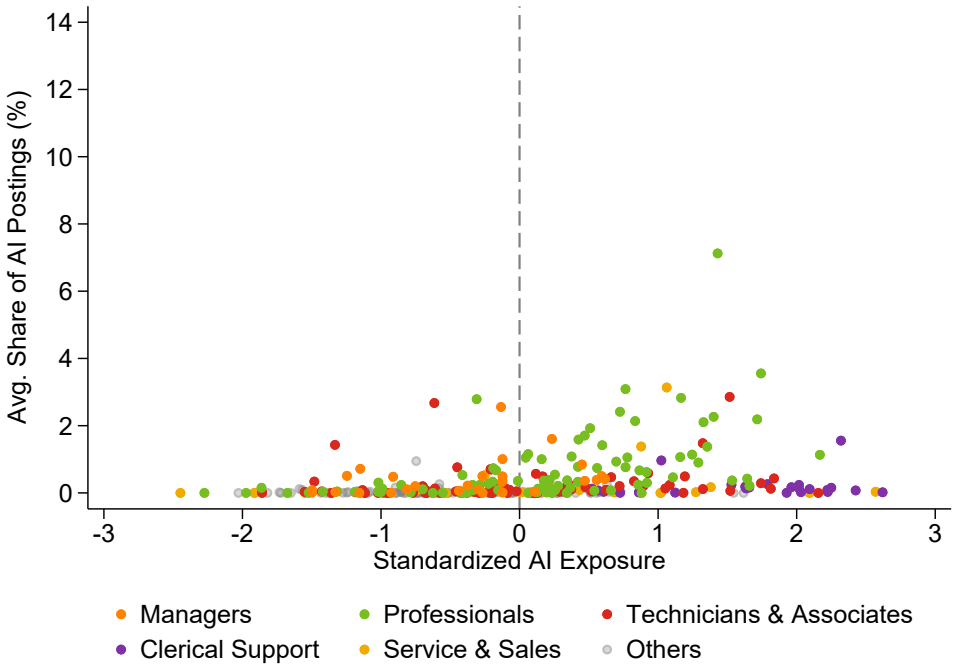
(a) November 2022 - October 2023



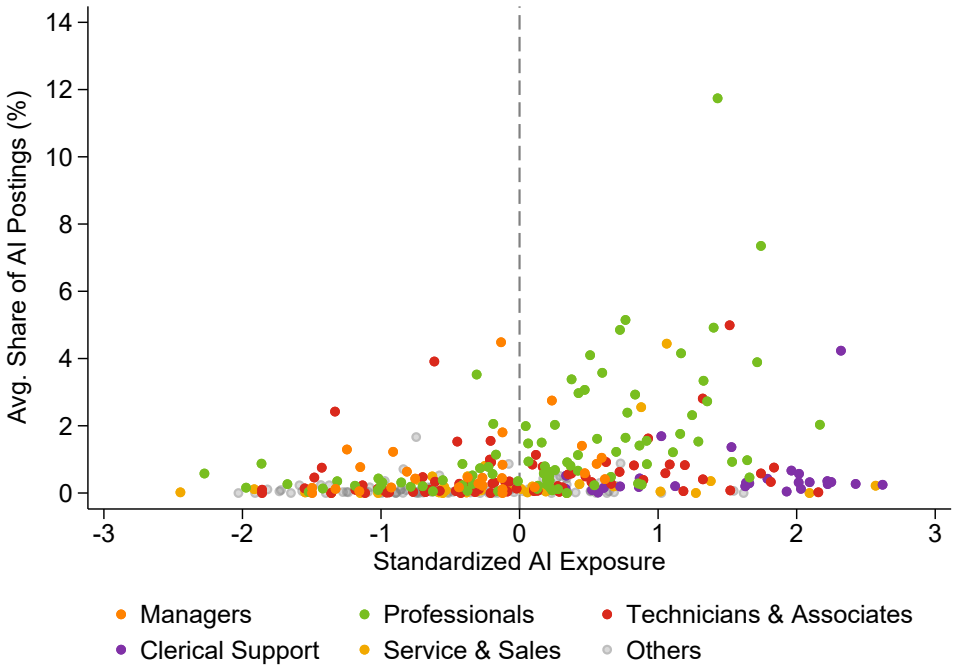
(b) November 2023 - October 2024

Notes: The sample comprises all economies. Share of AI postings is calculated as the share of postings that feature an emerging AI skill. Each dot represents a ISCO-08 4-digit occupation. Standardized AI Exposure Index is the z-score of the C-AIOE measure for each ISCO-08 4-digit occupation from Pizzinelli et al. (2023).

Figure C.3: Emerging AI skills over time (continued)



(c) November 2024 - October 2025

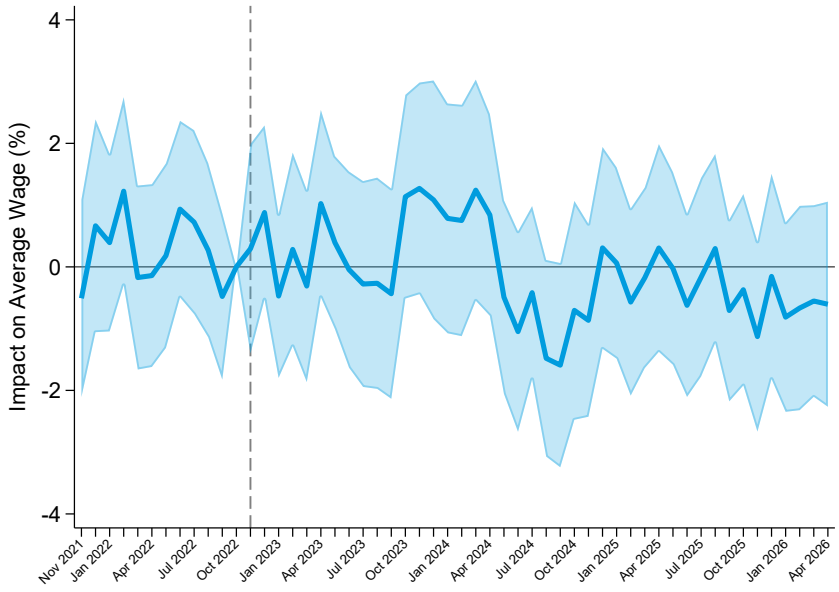


(d) November 2025 - April 2026

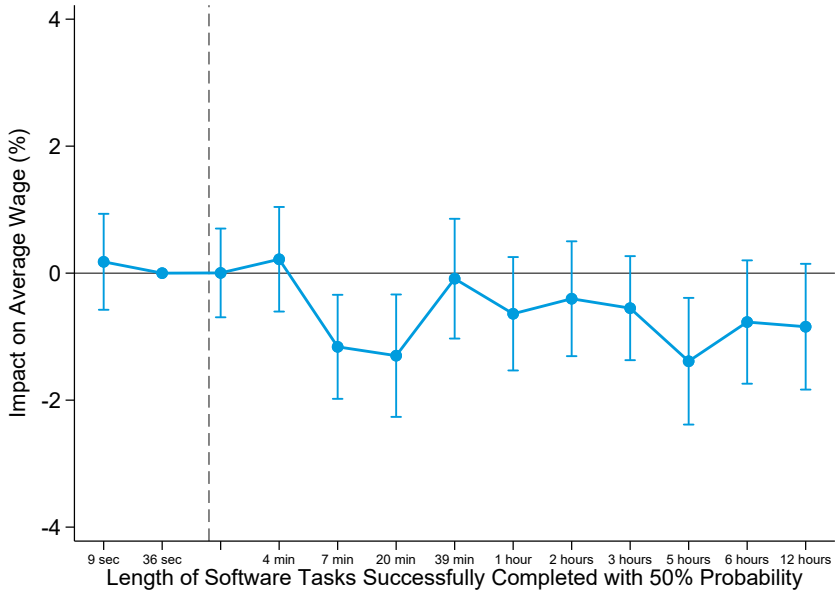
Notes: The sample comprises all economies. Share of AI postings is calculated as the share of postings that feature an emerging AI skill. Each dot represents a ISCO-08 4-digit occupation. Standardized AI Exposure Index is the z-score of the C-AIOE measure for each ISCO-08 4-digit occupation from Pizzinelli et al. (2023).

D Impact on wages

Figure D.1: No evidence of wage response so far (average wage)



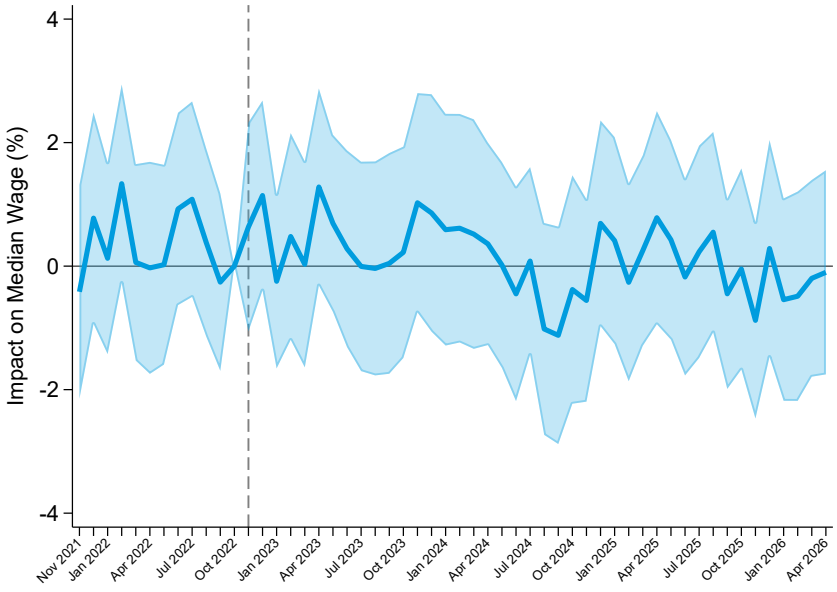
(a) Event study estimates



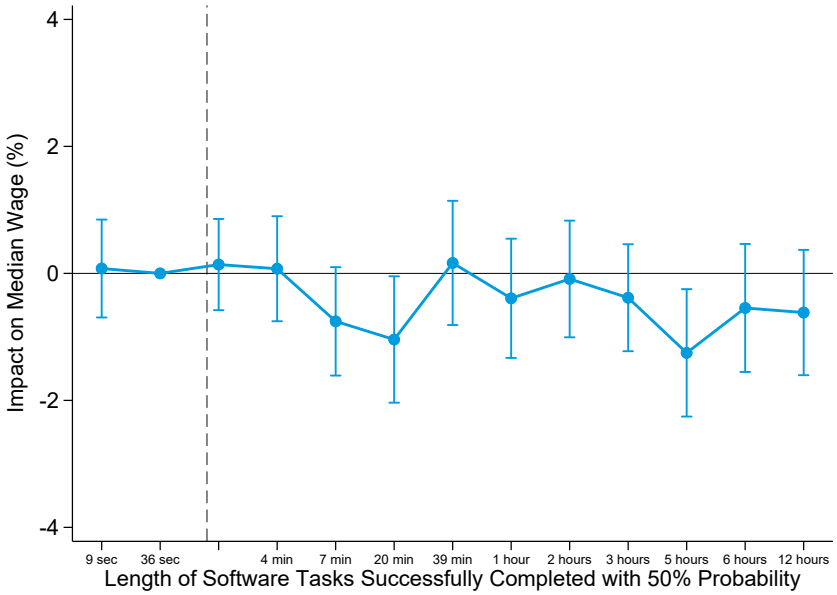
(b) Event study estimates and AI capability

Notes: The sample comprises all economies. Standard errors are clustered at the occupation level. Shaded bands and whiskers show 95 percent confidence intervals.

Figure D.2: No evidence of wage response so far (median wage)



(a) Event study estimates



(b) Event study estimates and AI capability

Notes: The sample comprises all economies. Standard errors are clustered at the occupation level. Shaded bands and whiskers show 95 percent confidence intervals.



PUBLICATIONS

How is Labor Demand Changing as AI Improves? New Global Evidence

Working Paper No. WP/2026/220